

A Wear Analysis and Age Estimation Model Based on Wear Depth Simulation

Summary

Staircases serve as silent witnesses to the rhythm of human life, bearing the imprints of countless footsteps that tell stories of daily passages. These wear patterns represent a complex interplay between human behavior and material response. To decode these physical signatures and provide scientific insights into stair wear mechanisms, we developed an integrated mathematical framework encompassing wear prediction, gait pattern detection and usage analysis.

For task 1, we proposed a **Dynamic Wear Depth Simulation Model**. By integrating the **Archard wear model** with human gait dynamics, we thoroughly analyzed the differences in wear rates during stair ascent and descent. Furthermore, we established a **footstep distribution model** for multiple users using a normal distribution and explored **the cumulative wear mechanism** over long-term usage. Through **Finite Element Analysis**, we identified the mechanism leading to more significant wear on stair edges and introduced a correction factor to improve accuracy.

For task 2, we developed a **Wear Characteristics Inverse Prediction Model** to investigate stair usage patterns. Based on the wear coefficients, we derived a formula for pedestrian flow to quantify stair usage frequency. Additionally, we proposed a **Directional Preference Index (DPI)** based on the **wear distribution** along the x -axis to analyze the user's inclination for ascending or descending the stairs. The model was also able to inversely predict the number of concurrent users on the stairs and assess their impact on different areas of stair wear.

For task 3, we established an **NDT Equipment selection Model**. Based on the need of two previous models, we determined the material parameters required for stair usage pattern analysis, including **step dimensions, wear depth, and hardness**. Subsequently, we conducted a comprehensive evaluation of non-destructive testing (NDT) methods and 17 NDT tools and applied the **Entropy Weight Method (EWM)** to evaluate six key criteria. With the **Cost-Benefit Analysis Decision Model** we decided the optimal detection tools and methods.

For Task 4, we developed a **Statistical Wear Consistency Model** to analyze the relationship between simulated and measured wear patterns. The model employs **Mean Squared Error (MSE)** to quantify wear consistency and utilizes **hypothesis testing with t-tests and p-value analysis** to validate stairwell age estimates. Additionally, **residual analysis** was integrated to detect anomalies and repair zones, while **ultrasound-based material testing** further ensured the credibility of material consistency evaluations.

For task 5, we developed a **Usage Pattern Inference Model**, integrating the findings from previous models. By applying **Maximum Likelihood Estimation (MLE)**, we derived **the time decay parameter α** and revealed its relationship with usage patterns: a larger α indicates a higher likelihood of the former pattern.

Finally, we validated our models' robustness through systematic wear rate sensitivity analysis and conducted a comprehensive assessment of their strengths and weaknesses.

Keywords: Stair wear, Archard wear model, Probability distribution, Mean Squared Error analysis, Age estimation.

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1 Introduction

1.1 Background

As a central element in architectural heritage, staircases preserve multidimensional archaeological insights. Their morphological transition—from primitive stone steps to contemporary composite material systems—serves as a direct reflection of sociocultural dynamics and technological advancements across civilizations. While existing studies predominantly determine construction chronology through material composition analysis, scant attention has been given to dynamic wear characteristics. Therefore, utilizing non-destructive techniques to collect data and developing mathematical models based on wear patterns could offer a methodological breakthrough in advancing archaeological research.

1.2 Restatement of the Problem

This study focuses on developing a mathematical model capable of predicting fundamental information about staircases based on their wear patterns. Specifically, the model aims to solve the following five tasks:

1. **Establish a model for stair wear mechanism.**
2. **Predict fundamental information about the stairs, including:**
 - The construction time of the staircase.
 - The preferred travel direction and travel patterns of users.
 - The general frequency and manner of staircase usage.
3. **Propose measurement methods to evaluate wear patterns, satisfying:**
 - The measurements must be non-destructive.
 - The cost, manpower, and tool requirements must be minimized.
4. **Validate the model's reliability and predictions, including:**
 - Assessing whether the predicted wear patterns align with the actual usage conditions of the stairs.
 - Determining the reliability of the predicted construction time.
 - Evaluating whether the wear patterns can accurately identify the material of the stairs.
5. **Extracting information regarding the frequency and style of usage.**

1.3 Literature Review

The methods to infer the age of ancient buildings by accessing the constant wear of stairs has long been an important topic. To achieve this, it is crucial to learn people's patterns ascending and descending stairs, and how common materials experience abrasion, especially those which are commonly used as a source material of stairs.

First it is necessary to learn the differences between ascending and descending stairs. [2] pointed out the mean foot overhang when descending stairs. It is also observed that there were

almost no foot overhang when ascending stairs. The wear patterns caused by the forefoot and heel on stair surfaces also differ significantly. [1] showed that forefoot tends to cause greater wear on stair treads. Moreover, [4] provides a more detailed overview on stair ascent and descent at different inclinations. [3] examines the required ground reaction force for each stair material for both ascending and descending stairs, which is important to the model. Therefore, we have incorporated these factors into our model.

One classic paper discussing material wear is [5], which established the classic Archard wear model, and also very widely used and further extended. Based on this, a linear model similar to Archard wear model is also adapted.

2 Assumptions and Justification

- **Differences among different individuals can be ignored.** Assume that the way each person's feet contact the stairs is the similar, and they have similar preferences when ascending or descending stairs. In particular, the differences among people's foot size, contact force, sliding speed, the position to put their feet on, etc., can be simplified as a probability distribution.
- **The material of the stairs is uniform** (if the stairs had been mended, assume that each type of material is uniform). To simplify calculations, this assumption ensures that abrasion occurs uniformly.
- **The wear of stairs may transform a flat surface into a slope** Here we ignore the possible influences on the wear rate due to the changes in the shape.
- **Ignore the impact of weather conditions**, which might slightly accelerate the constant wear of stairs. Here we assume that anthropogenic actions have a more significant impact on the constant wear of stairs, compared to weather such as rainfall, sandstorm, etc..

3 Notations

Symbol	Definition	Unit
Q	pedestrian flow rate	people/year
N	number of people travelling simultaneously	people
k	wear rate of foot while ascending	mm/people
k_f	wear rate of front of foot while descending	mm/people
k_b	wear rate of back of foot while descending	mm/people
Δx	average distance of people on the same stair	mm
λ	proportion of people going upstairs	-
w	width of stairs	mm
h	riser of stairs	mm
d	thread depth of stairs	mm
l_f	average foot length	mm
w_f	average foot width	mm
$(x_l, y_l), (x_r, y_r)$	center of left/right foot	mm
(x_c, y_c)	midpoint of center of left and right foot	mm
$S(x, y)$	longitudinal depth of wear per person	mm/person

Table 1: Notations

4 Our Work

According to the requirements, our work can be shown in the following figure.

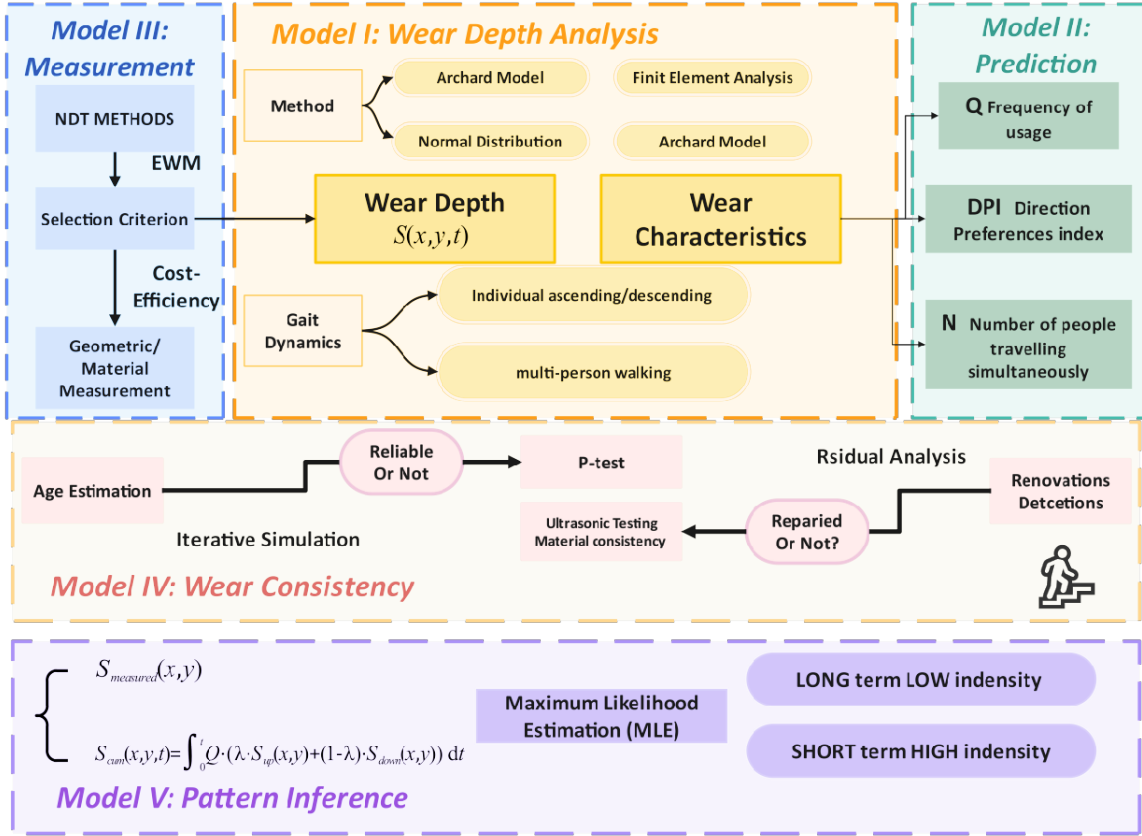


Figure 1: Our Work

5 Model I: Dynamic Wear Depth Simulation Model

To establish a dynamic model that can systematically analyze the wear mechanism of stairs, we first focus on the wear mechanism of a single person ascending or descending stairs by establishing a functional expression for the wear depth $S(x, y)$. Then the complexity of group behavior is introduced to analyze the spatial distribution effect of different path choices on the wear location when multiple people walk up and down the stairs in parallel. Finally, the cumulative effect of long-term repeated stepping is superimposed on this model.

5.1 Individual Footstep Distribution and Wear Mechanism

To clearly represent the wear positions and depths, a two-dimensional depth matrix is established for the stair tread. The length of the stair tread is defined as the x -axis, while the width is defined as the y -axis. The wear depth of the stair surface at point (x, y) is represented by the function $S(x, y)$.

We define the midpoint of the left and right foot centers as x_c . In the case of single-person traffic, due to the randomness and centralization characteristics of human gait during stair ascent and descent, the distribution of x_c follows a normal distribution:

$$x_c \sim \mathcal{N}(\mu, \sigma),$$

where μ represents the central position of foot placement, and σ is the standard deviation that characterizes the dispersion of foot positions.

To describe the positions of the left and right foot centers during stepping, we represented the center of left foot as (x_l, y_l) and the center of right foot center as (x_r, y_r) . Thus the longitudinal position can be expressed as:

$$x_l = x_c - \frac{d_1}{2}, x_r = x_c + \frac{d_1}{2},$$

where d_1 denotes the distance between the centers of the left and right feet.

The lateral positions of both feet are assumed to be the same during typical stair use, since the feet are aligned along a single walking path in the longitudinal direction (the x -axis). As a result, we can assume that the center of the left and right feet positions along the y -axis are identical, denoted as y_f .

While calculating the exact position of the feet along the y -axis, we should take the phenomenon of “foot overhang” which means when a person steps on a stair tread, their foot typically extends slightly beyond the edge of the step. The lateral position y_f is determined by the geometric projection of the foot onto the stair tread and can be calculated as:

$$y_f = l_f \cdot (0.5 - p),$$

where p represents the proportion of the foot length that overhangs the stair. After conducting an in-depth literature study and analyzing the data from 113 groups [2], we can conclude that the average foot overhang rate for adults is $p = 17.07\%$.

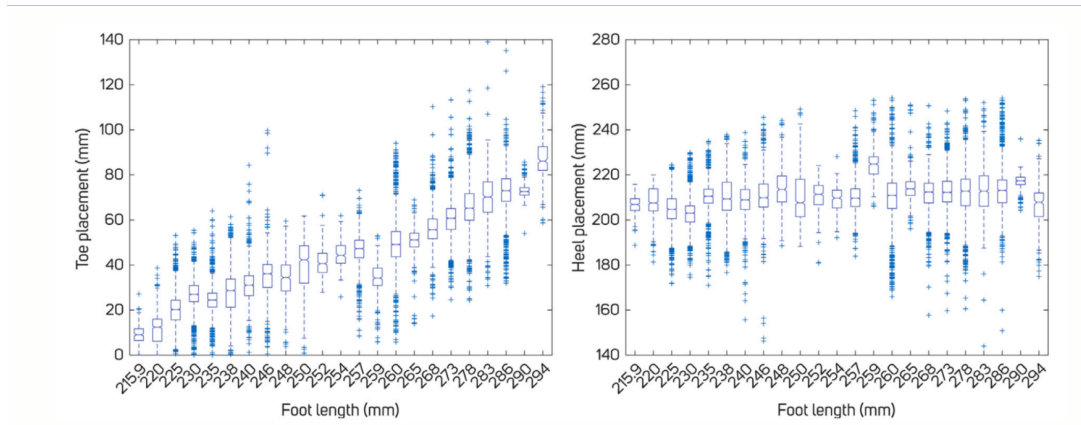


Figure 2: Toe and heel placement: distance measured from the edge.

Once the center positions (x_{cl}, y_{cl}) and (x_{cr}, y_{cr}) are established, the boundaries of the wear zones can be geometrically defined. These zones represent the contact area between the foot and the stair surface, calculated as:

$$x \in [x_l - \frac{w_f}{2}, x_r + \frac{w_f}{2}], \quad y \in [y_f - \frac{l_f}{2}, y_f + \frac{l_f}{2}]. \quad (1)$$

After completing the geometric modeling of the foot, we proceed to establish the wear depth model. According to the literature research, the trajectory of the center of pressure (CoP) differs

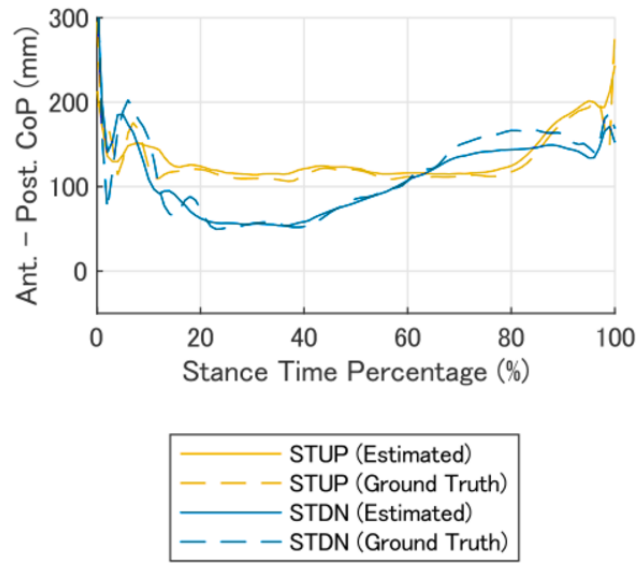


Figure 3: Waveforms of CoP on the anterior-posterior direction.

significantly between the process of ascending and descending stairs, which requires separate analyses of the wear characteristics in these two scenarios.

Ascending Phase:

During stair ascent, as supported by the findings from the study, we can find that the center of pressure (CoP) remains nearly stable along the medial-lateral (x -axis) direction and only exhibits minor fluctuations along the anterior-posterior (y -axis) direction, which consistently concentrated in the central region. This indicates that the pressure distribution during the ascent phase is approximately uniform. Thus based on the analysis, the wear rate across the entire foot contact area can be approximated as constant, represented by k . And the single-step wear depth during the ascent denoted as S_{up} can be expressed as

$$S_{up}(x, y) = k \cdot \exp\left(-\frac{(x - \mu_i)^2}{2\sigma^2}\right) \quad (2)$$

Descending Phase:

During stair descent, however, we observe that the center of pressure (COP) exhibits significant fluctuations along the anterior-posterior direction, primarily concentrated in the toe region. This indicates that the pressure distribution during descent is highly uneven, with greater pressure in the toe region compared to the rear of the foot. Such a distribution makes the edges of the step more prone to wear. Based on experimental data from the literature, we define the wear rate of the toe region as k_f and the wear rate of the rear foot as k_b . With the previously established foot model, the wear depth can be expressed as

$$S_{down}(x, y) = \begin{cases} k_f \cdot \exp\left(-\frac{(x - \mu_i)^2}{2\sigma^2}\right) & y \in \left[y_f - \frac{l_f}{2}, y_f\right], \\ k_b \cdot \exp\left(-\frac{(x - \mu_i)^2}{2\sigma^2}\right) & y \in \left[y_f, y_f + \frac{l_f}{2}\right]. \end{cases}$$

The wear-depth model for individual taking the single step is established.

5.2 Multi-Person Dynamics Analysis

When a group of people ascend or descend the stairs simultaneously, the symmetric normal distribution for the center point of footsteps on the x -axis, as used in the single-person model, is no longer valid. This is because different pedestrians occupy distinct portions of the stair width, which lead to a more complex lateral distribution of footsteps. Consequently, it is necessary to revise the distribution model from the single-person scenario to accurately reflect the distribution of footsteps when people walk side by side.

In this case, we still use the center of the stair width as the reference point for generating the footstep center points. We assume that there are N people walking up or down the stairs at the same time, then the footstep center point of each people, denoted as μ_i , can be expressed as

$$\begin{aligned}\Delta_{\text{offset}} &= \frac{N-1}{2} \cdot \Delta x \\ \mu_i &= \frac{w}{2} - \Delta_{\text{offset}} + i \cdot \Delta x, \quad i = 0, 1, \dots, N-1,\end{aligned}\quad (3)$$

where Δx denotes average distance of people on the same stair, and w denotes the width of the stair.

Also the standard deviation of the normal distribution needs to be redefined. When the crowd density is high, the individual footstep distribution range is compressed due to spatial constraints, causing the standard deviation σ_x to decrease. Conversely, when the crowd is sparse, the standard deviation σ_x gradually approaches the single-person walking standard deviation σ_{max} . Based on this characteristic, the relationship between σ_x for multiple people walking side by side and σ_{max} can be expressed as:

$$\sigma_x = \sigma_{\text{max}} \cdot \sqrt{1 - \frac{Nw}{w}}.$$

Here $Nw = N \cdot \Delta x$ denotes the total width occupied by the crowd, representing the horizontal crowd density on the stairs, and σ_{max} denotes the standard deviation of the normal distribution in the individual ascent/descent model.

Under the condition of multiple people walking side by side, the distribution of the center points of footsteps along the x -axis needs to be updated as:

$$x_{\text{ci}} \sim \mathcal{N}(\mu_i, \sigma_x).$$

5.3 Cumulative Wear Analysis over Time

Then we need to further extend our model to evaluate the long-term impact of pedestrian traffic on stair treads. The cumulative wear is directly influenced by the pedestrian flow, walking direction, and material properties.

The cumulative wear depth at position (x, y) over time t can be defined as:

$$S_{\text{cum}}(x, y, t) = \int_0^t Q \cdot (\lambda \cdot S_{\text{up}}(x, y) + (1 - \lambda) \cdot S_{\text{down}}(x, y)) \, dt, \quad (4)$$

where Q represents the Pedestrian flow rate (people/year),

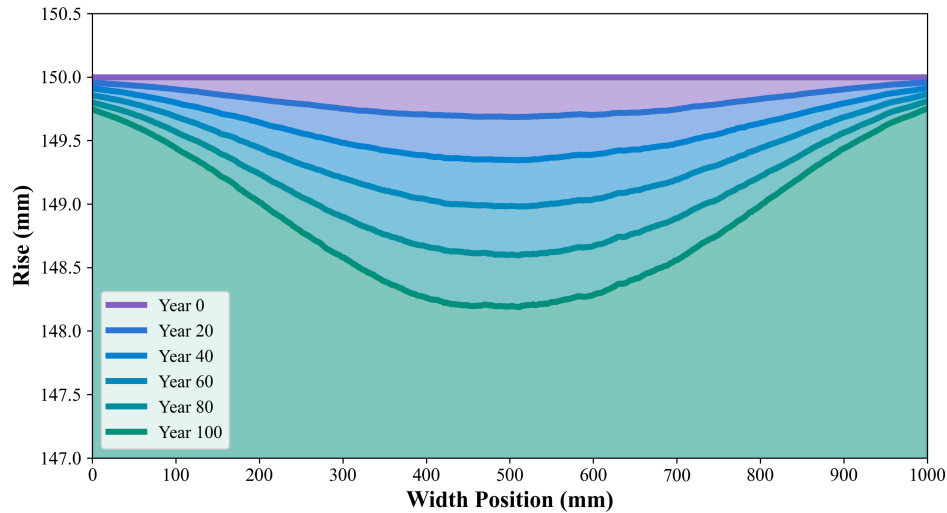


Figure 4: The cumulative wear effect

As shown in Figure 4, we can identify the cumulative effect of wear over time,

After comparing the actual wear patterns with the predicted results from our model, we found that stair edges experience significantly more wear, likely due to the lack of supporting structures.

To verify this, we conducted **finite element analysis (FEA)** using ANSYS, modeling a single person's load on the stairs, including a density of $7.85 \times 10^{-6} \text{ kg/mm}^3$, Young's modulus of 200 GPa, and Poisson's ratio of 0.3. A 1000 N load was applied to the stair tread (width: 1.2 m, length: 0.3 m), with appropriate boundary constraints to simulate wall support. The result is shown in Figure 5.

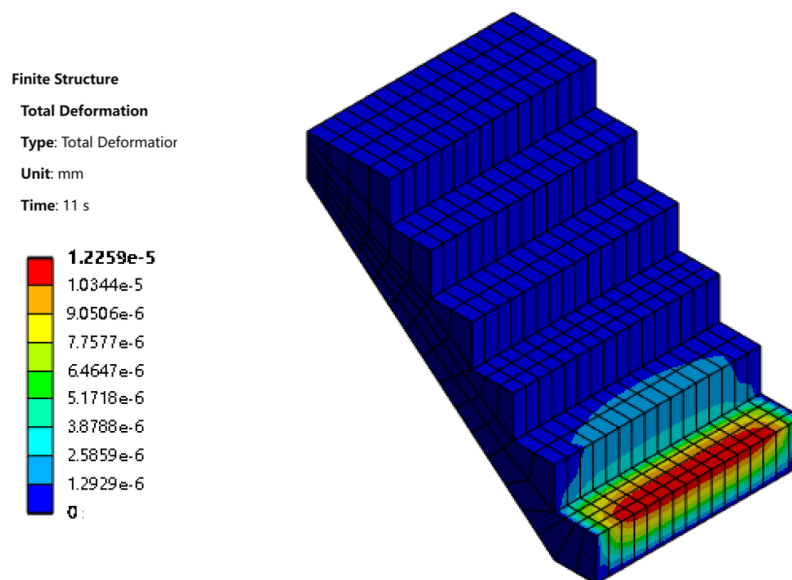


Figure 5: The result of FEA ANSYS simulation

Furthermore, the refined wear depth noted $S'(x, y, t)$ is expressed as:

$$S'(x, y, t) = S_{\text{cum}}(x, y, t) - (h_0 - S_{\text{cum}}(x, y, t)) \cdot e^{-\alpha t}. \quad (5)$$

5.4 Simulation Analysis and Model Validation

Based on the model we established, we developed a simulation tool using Python. By setting parameters such as pedestrian flow rate Q , directional flow ratio λ , simulation duration t , and material properties of the staircase, the simulator can generate wear distribution under various conditions. Using this tool, we can compare the simulation results with reference data from existing literature to validate the accuracy and reliability of our model.

Given the limited availability of publicly accessible data on staircase wear, we selected a representative real-world case to verify the correctness of our model. A detailed analysis of the validation results is presented below.

Table 2: Key Parameters for the stair case

Parameter Name	Value
Stair Dimensions	Width: 1500 mm, Tread: 300 mm, Rise: 150 mm
Annual Traffic	80000 people
Years of Use	50 years
Single (middle) Ratio	0.1
Group (left/right) Ratio	0.9
Left-Up/Right-Down Ratio	0.2
Right-Up/Left-Down Ratio	0.8

The heat-map (fig. 6) generated from the simulation clearly shows the area and shape of the worn regions, which closely resemble the actual wear pattern. The deepest wear is observed at the edges of the steps, with a depth of approximately 1.03mm. Additionally, the wear distribution is noticeably uneven between the left and right sides, indicating the presence of parallel two-person traffic on the staircase. Specifically, upward traffic tends to favor the right side, while downward traffic tends to favor the left side, aligning with common preferences for directional movement.

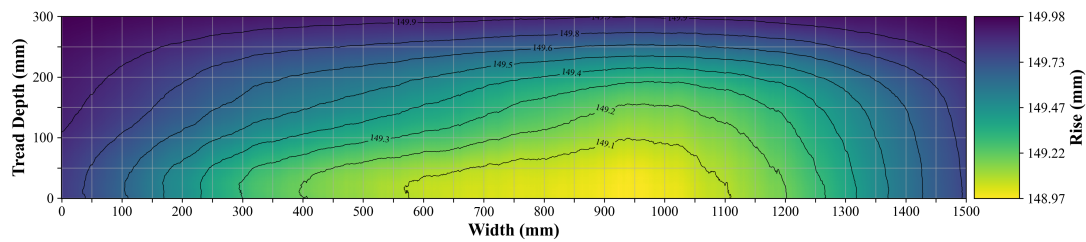


Figure 6: Uneven wear distribution between left and right sides.

6 Model II: Wear Characteristics Inverse Prediction Model

After establishing the relationship between wear depth, location, and time, we can develop an inverse prediction model based on that relationship. This model should allow us to use the data on stair wear to infer the stairs' historical usage, including the usage frequency, users' directional preferences and patterns of parallel multi-person traffic.

6.1 Usage Frequency

By estimating the pedestrian flow rate Q , usage frequency of stairs can be calculated: Measure the average depth of wear and the area of width, we can obtain the total volume of wear V . Given the wear coefficient and hardness of the material, we can calculate the value of Q .

The total volume of wear V can be calculated as:

$$V = \iint S'(x, y, t) dx dy$$

$$\approx Qt(\lambda k + (1 - \lambda)(k_f + k_b)/2)l_f w_f,$$

from which we immediately obtain $Q = \frac{V}{tl_f w_f(\lambda k + (1 - \lambda)(k_f + k_b)/2)}$.

6.2 User's Directional Preferences

By calculating the wear characteristics in the longitudinal direction (y -axis) of the stair tread, it is effectively determined whether the stairs were predominantly used for ascending or descending. As shown in Figure 7, we can identify the differences the wear depth distribution between the ascending and descending phases.

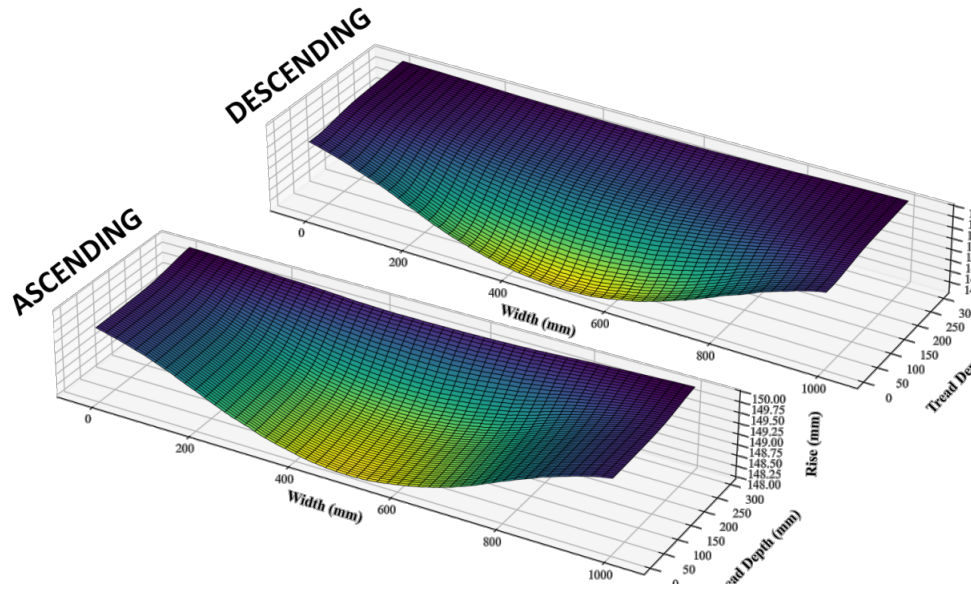


Figure 7: The wear depth simulation of single person

We can obtain the wear depth at each point $S_r(x, y)$ on the stair tread by performing a three-dimensional modeling of the stair tread surface on-site.

The stair tread is divided into two regions: the front region (F), which is the front half of the stair tread ($y \in [y_c, l]$, where y_c is the y -coordinate of midpoint), and the rear region (R), which is the rear half ($y \in [0, y_c]$).

The average wear depth in the front and rear regions is calculated as follows:

$$W_F = \frac{1}{A_F} \iint_F S(x, y, t) dx dy, \quad W_R = \frac{1}{A_R} \iint_R S(x, y, t) dx dy,$$

where A_F and A_R represents the area of the front and back region respectively.

The wear depth ratio R_W is then defined as $R_W = \frac{W_F}{W_R}$, where $R_W > 1$ indicates a descending preference and $R_W < 1$ indicates an ascending preference.

To further quantify the intensity of directional preference, we introduce a Directional Preference Index (DPI), defined as:

$$\text{DPI} = \alpha(R_W - 1),$$

where α is a weighting coefficient to adjust the sensitivity of the model to R_W . A positive DPI value indicates a descending preference, while a negative value indicates an ascending preference. The magnitude of DPI represents the strength of the directional preference.

6.3 Patterns of Multi-Person Traffic

We need to incorporate the characteristics of footstep distribution during multi-person walking and the superposition relationship of the overall wear area mentioned in Model I. By combining the staircase width with the measured position and size of the wear area, we can estimate the number of people walking side by side on the staircase N .

Along the x -axis, The wear depth $S(x)$ distribution is represented as the superposition of all individual footsteps as:

$$S(x) = \sum_{i=1}^N [\lambda \cdot S_{\text{up},i}(x) + (1 - \lambda) \cdot S_{\text{down},i}(x)].$$

$S_{\text{up},i}(x)$ is shown in Equation (2) and $S_{\text{down},i}(x)$ is shown in Equation (1) represent the wear depth distributions for the i -th person during ascent and descent, respectively.

We should also notice that each footstep center positions μ_i are distributed as shown in Equation (3).

Following the geometric constraints, the staircase width w limits the lateral distribution of footsteps. Specifically, the footstep spacing Δx and the total number of footsteps N must satisfy:

$$N \cdot \Delta x \leq w.$$

This condition ensures that all footsteps and wear areas remain within the staircase boundaries.

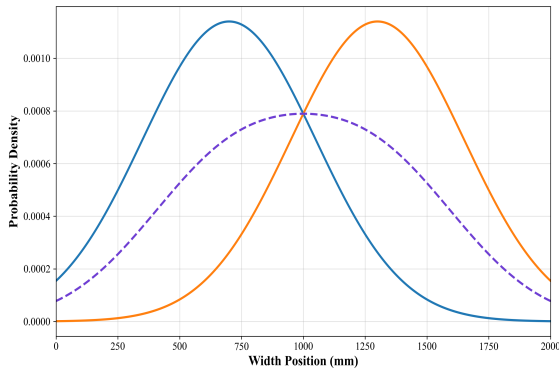


Figure 8: Foot step distribution when $N = 2$

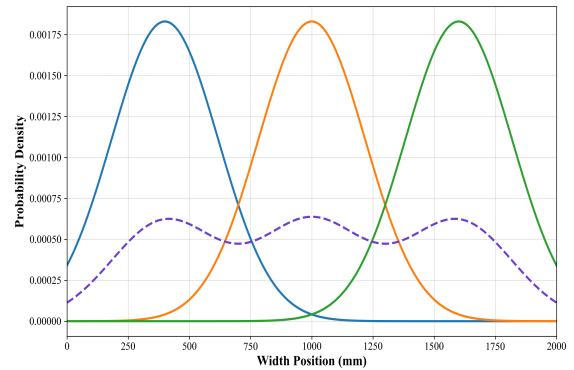


Figure 9: Foot step distribution when $N = 3$

By analyzing the wear depth distribution across the stairs, each peak in the distribution can be identified as representing the center of a footstep. Also with the simulation, we can plot the

distribution of the footstep in the cases with different numbers of people walking on the same stairs. Figure 8 and 9 verifies the analysis that each peak can represent the center of a foot step.

By constructing a three-dimensional model of the stairs to obtain the wear depth distribution with the help of NDT tools selected, we can perform peak detection on the resulting data. However, the complexity of the wear mechanisms in real-world scenarios may make the raw data challenging to analyze. To address this, **Gaussian filtering** can be applied as a preprocessing step to smooth the raw data, reducing noise and irregularities. This allows for more accurate identification of all peaks in the wear depth distribution. With the smoothed wear depth distribution, we can obtain the numbers of peaks in the distribution. In this way, the estimated people (N) walking on the same stairs simultaneously is identical to the numbers of peaks we observe in the wear depth distribution.

7 Model III: Non-Destructive Testing Equipment Selection Model

To obtain the stair-related parameters required for the completion of the study without damaging the structural integrity of the stair, the following two types of parameters need to be obtained by **Non-destructive testing** (NDT) techniques in this study:

Category	Parameter and Symbol
Geometric Data	Depth of wear on the stair surface (Δh)
	Width of wear (W)
	Original dimensions ($w \times h \times d$)
Material Parameters	Stone hardness (H)
	Wear coefficient (k)

Table 3: Geometric Data and Material Parameters

To address these needs, a **cost-benefit evaluation model** is established. This model evaluates the equipment by analyzing its performance (benefit) and cost metrics, ensuring that the selected equipment offers the highest cost-efficiency. The cost-benefit approach directly aligns with the key requirements of the task: minimizing financial costs while ensuring the practicality and effectiveness of the equipment.

We select 17 representative types of NDT equipment that meet the GB/T 34370-2017 General Technical Requirements for Non-Destructive Testing Instruments, denoted by a_i , where $i = 1, 2, \dots, 17$. According to the archaeological site's actual needs, the NDT equipment is divided into two main categories: geometric data acquisition equipment (category G) and material properties testing equipment (category M). And all the related NDT methods are shown in Figure 7.

Referring to the specifications for the NDT techniques in archaeological sites and adhering to the principles of minimizing equipment and labor costs, six evaluation indicators b_j , $j = 1, 2, \dots, 6$, are established:

1. Performance Metrics (Benefit Indicators):

- **Measurement Precision:** The accuracy of the data collected.

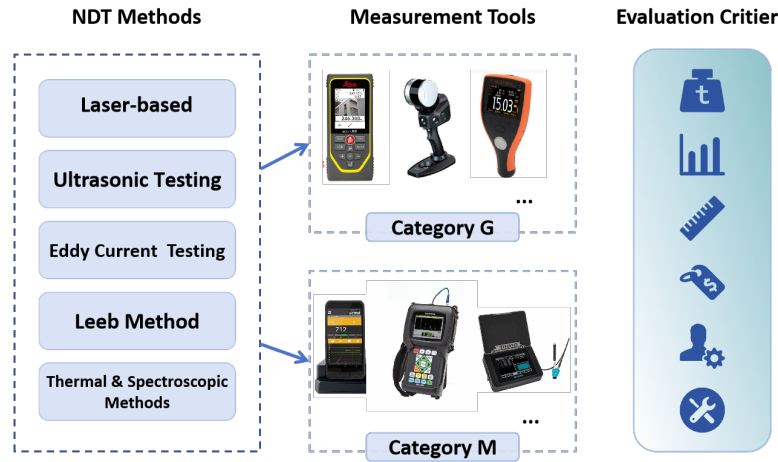


Figure 10: The NDT Methods and tools

- **Portability:** Adaptability to confined spaces.
- **Data Collection Efficiency:** The ability to collect multiple parameters simultaneously.

2. Cost Metrics (Cost Indicators):

- **Purchase Cost:** Initial investment required for the equipment.
- **Maintenance Cost:** Annual servicing costs.
- **Operational Cost:** Per-operation cost in terms of manpower and resources.

The entropy weight method (EWM) is applied to determine the weight of each indicator.

Weight determination

Let $X = (x_{ij})_{n \times m}$ be the original decision matrix, where x_{ij} denotes the value of the i -th equipment under the j -th criterion.

With the standardized matrix Z_0 , The information entropy e_j for the j -th criterion is calculated as follows:

$$e_j = \frac{1}{\ln(n)} \sum_{i=1}^{17} p_{ij} \ln(p_{ij}), \quad p_{ij} = \frac{r_{ij}}{\sum_{i=1}^{17} r_{ij}},$$

where p_{ij} indicates the proportion of the i -th equipment under the j -th criterion.

The weight w_j for the j -th criterion is determined using the formula:

$$w_j = \frac{1 - e_j}{\sum_{j=1}^m (1 - e_j)},$$

where w_j indicates The normalized weight for the j -th criterion.

Respectively, we calculate the criterion weight for the G-category and M-category. Using the calculated weights, each piece of equipment is evaluated based on its **benefit score** (E_i) and **cost score** (C_i). These scores quantify the overall performance and economic feasibility:

- **Benefit Score (E_i):** Reflects the weighted sum of benefit indicators, including precision, portability, and efficiency.
- **Cost Score (C_i):** Reflects the weighted sum of cost indicators, including purchase, maintenance, and operational costs.

The **Relative Evaluation Coefficient (REC)** is then computed as the ratio of benefit to cost:

$$REC_i = \frac{E_i}{C_i}.$$

Higher REC_i values indicate better cost-effectiveness.

Based on the evaluation model, the REC scores for all equipment are ranked to determine the most suitable options for both Category G and Category M. In Figure 11, we present the performance of the top three devices with the highest REC scores in each category across various metrics. It is important to note that for cost-related metrics, lower values are preferable.

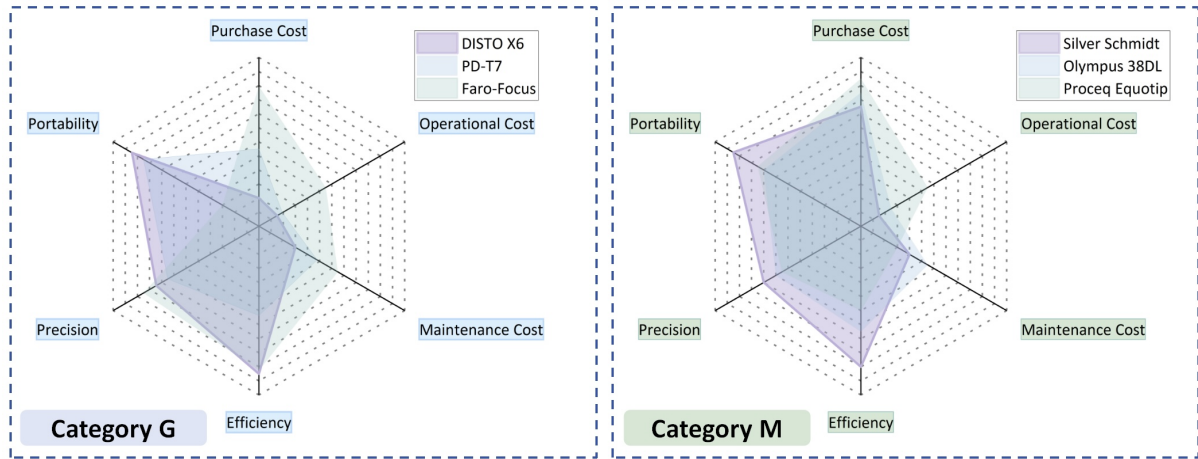


Figure 11: The performance of the top three devices

The result helps us conclude with the most recommended measurement plan assuring both the minimum cost and maximum flexibility meanwhile obtaining all the parameters we need. Following is a brief introduction of the devices included in the measurement plan:

- **Leica DISTO X6:** Utilizes **Laser-based NDT Method** for precise geometric measurements. It employs pulsed laser technology to capture stair wear width (W) and original dimensions ($w \times h \times d$) in confined spaces, achieving ± 1.0 mm accuracy.
- **Proceq SilverSchmidt:** Applies **Rebound Hardness Testing (Leeb Method)** to evaluate stone hardness (H). By measuring the rebound velocity of a spring-driven impact probe, it enables rapid, non-destructive hardness assessments in narrow environments.
- **Olympus 38DL PLUS:** Implements **Ultrasonic Testing (UT)** with high-frequency sound waves to analyze material properties. This method calculates thickness and correlates acoustic velocity with density for wear coefficient (k) determination at ± 0.1 mm precision.

8 Model IV: Statistical Wear Consistency Model

The wear patterns on a stairwell are an essential indicator of its historical usage, the materials used, and potential renovations or repairs over time. Archaeologists often face challenges in validating whether the wear observed aligns with historical records or simulations. In this section, we propose a statistical model that evaluates the consistency of the wear patterns with the provided information, estimates the age of the stairwell, detects any renovations, and determines whether the source material aligns with the observed wear.

8.1 Wear Consistency Analysis

Since wear patterns on a stairwell offer valuable insights into its historical usage, this subsection proposes a statistical framework to evaluate their consistency with simulated predictions.

Model Development

To evaluate wear consistency, we compare the **measured height matrix** H , derived from sampling the actual stairwell, with the **simulated height matrix** $\hat{H}$, generated based on the given assumptions and our previous model. Since H cannot be fully obtained due to measurement constraints, the comparison is performed on representative **sampled data vectors** extracted from both H and $\hat{H}$.

The sampled data vector, denoted as $\mathbf{h}$, is constructed by identifying the primary worn region of each tread and uniformly sampling 50 points within this region. The measured height at each sampling point is recorded using a portable laser rangefinder. Similarly, the corresponding vector $\hat{\mathbf{h}}$ is derived from the same locations in the simulated matrix $\hat{H}$.

To quantify the similarity between $\mathbf{h}$ and $\hat{\mathbf{h}}$, we employ the **Mean Squared Error (MSE)**. MSE compares the squared differences between the two vectors:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \left(h_i - \hat{h}_i \right)^2,$$

where h_i and $\hat{h}_i$ are the sampled heights from H and $\hat{H}$, respectively, and N is the number of sampled points (here $N = 50$).

A smaller MSE value indicates higher similarity between the measured and simulated wear patterns. By directly using the squared differences without normalization, the metric provides an absolute measure of the deviation between the two vectors, making it sensitive to differences in magnitude.

By sampling H and comparing it to $\hat{H}$ at equivalent locations, the model provides a robust assessment of wear consistency while accounting for the practical limitations of direct measurement.

Model Validation

To validate the proposed model for wear consistency analysis, we utilize two artificially generated datasets. The first dataset, **Sample 1**, represents a *consistent sample* by introducing Gaussian noise to a simulated 100-year wear pattern. The second dataset, **Sample 2**, serves as an *inconsistent sample* by applying the same noise to a simulated 120-year wear pattern. Both datasets are compared against the simulated wear matrix for 100 years $\hat{H}$ under the same

conditions.

1. Sampling Region and Points

The selected region for analysis, along with the 50 uniformly sampled points, is shown in Figure 12. The selected region is highlighted in red, and sampled points are indicated with black dots. This region captures the primary worn area on the stair treads, ensuring representative coverage of wear patterns.

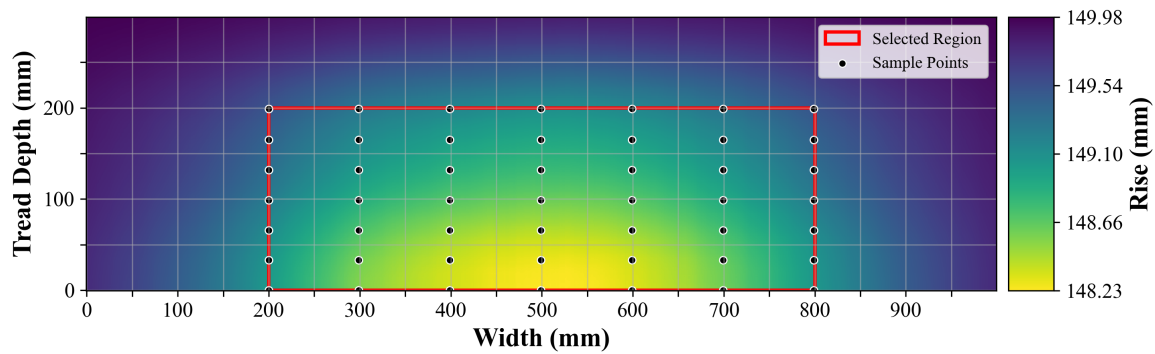


Figure 12: Sampling Region and Points

2. Residual Analysis

Figures 13 and 14 depict the residuals $h_i - \hat{h}_i$ for Sample 1 and Sample 2, respectively. The residuals are plotted as stem plots to highlight deviations across the 50 sampled points. Sample 1 shows minor deviations, reflecting its consistency with the 100-year prediction, while Sample 2 exhibits significant deviations due to its mismatch with the assumed age.

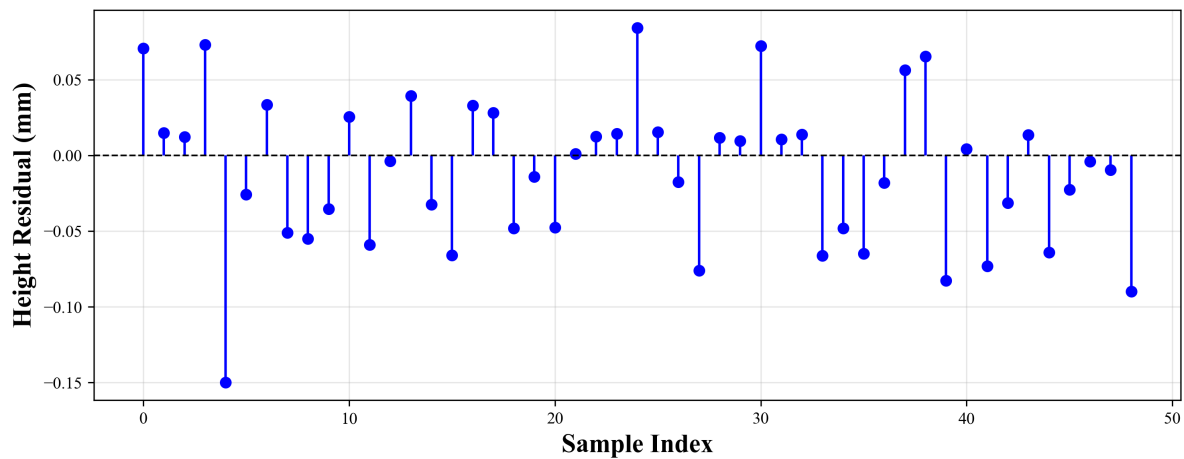


Figure 13: Residuals between Measured and Simulated Heights for Sample 1

3. Scatter Plot of Measured vs. Simulated Heights

Figure 15 compares the measured heights h with the simulated heights $\hat{h}$ for both samples. The $y = x$ line represents perfect agreement. Sample 1 points are closer to this line, while Sample 2 shows a larger spread, further confirming its inconsistency.

4. Mean Squared Error (MSE)

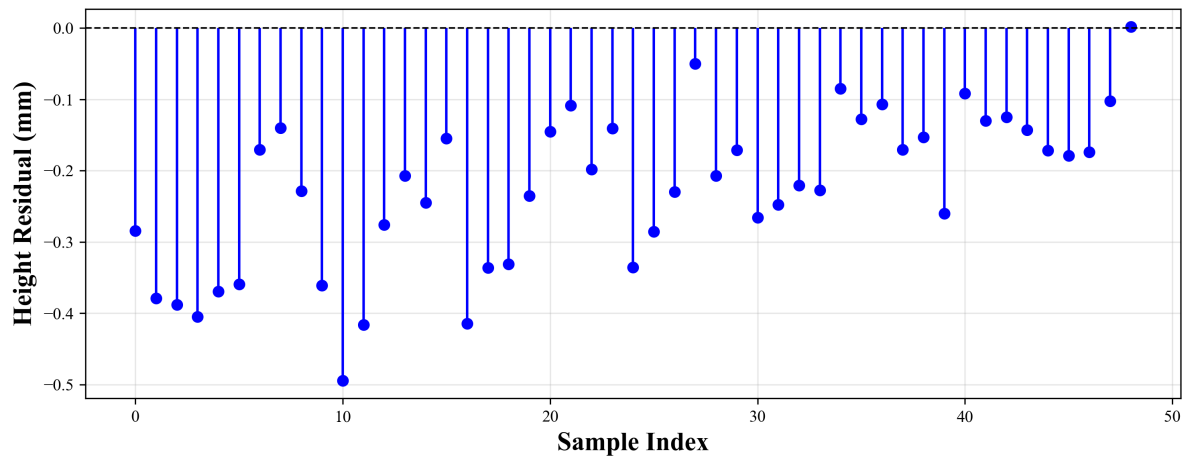


Figure 14: Residuals between Measured and Simulated Heights for Sample 2

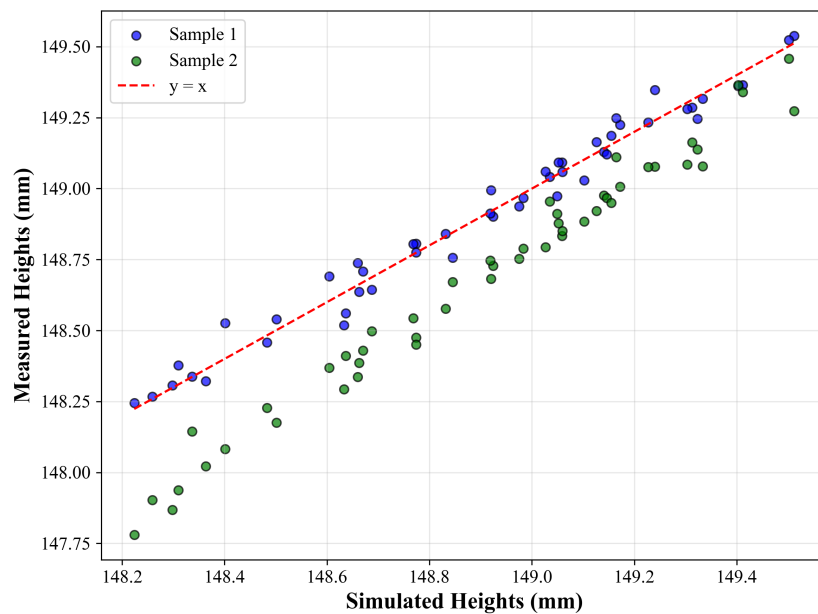


Figure 15: Scatter Plot of Measured vs. Simulated Heights for Sample 1 and Sample 2

The MSE metric quantifies the similarity between the measured h and simulated $\hat{h}$ wear patterns.

- **Sample 1:** $\text{MSE} = 0.001723$.
- **Sample 2:** $\text{MSE} = 0.062027$.

The smaller MSE for Sample 1 aligns with its consistency, while the larger value for Sample 2 confirms its deviation from the predicted wear pattern.

8.2 Age Estimation and Reliability

Since accurate age estimation is vital in archaeological studies, this subsection proposes a model to estimate the age of the stairwell by comparing measured maximum wear depths

with predicted values. Additionally, the reliability of the estimated age is evaluated through **hypothesis testing**.

Model Development

1. Model Preparation

Using our **Model I**, the stairwell's wear pattern is simulated based on daily traffic volume, and directional flow. The simulated wear matrix $\hat{W}(t)$ represents the wear depths across different locations on the stair treads in year t . From this matrix, the predicted maximum wear depth for a given age is denoted as:

$$\hat{w}_{\max}(t) = \max(\hat{W}(t)).$$

For the measured data, a series of stair treads are selected as the sample. The maximum wear depths, represented by $W_1, W_2, \dots, W_n$, are measured using non-destructive tools, such as a laser rangefinder. These measurements yield the corresponding sample values $w_1, w_2, \dots, w_n$. The true underlying population of maximum wear depths is denoted as W , with a population mean μ . The sample mean statistic $\bar{W}$ and its calculated value $\bar{w}$ are expressed as:

$$\bar{W} = \frac{1}{n} \sum_{i=1}^n W_i, \quad \bar{w} = \frac{1}{n} \sum_{i=1}^n w_i.$$

2. Age Estimation via Iterative Simulation

To estimate the stairwell's age, the measured sample mean $\bar{w}$ is compared iteratively with predicted maximum wear depths $\hat{w}_{\max}(t)$ simulated for various ages t . By minimizing the absolute difference:

$$\Delta(t) = |\bar{w} - \hat{w}_{\max}(t)|,$$

the estimated age t_{best} is identified as the year for which the simulated wear depth most closely aligns with the measured wear depth:

$$t_{\text{best}} = \arg \min_t \Delta(t).$$

3. Reliability Assessment

To evaluate the reliability of the estimated age, a **one-sample p-test** is conducted. The null hypothesis H_0 asserts that the mean of the measured maximum wear depths matches the predicted maximum wear depth at the estimated age $\hat{t}$:

$$H_0 : \mu = \hat{w}_{\max}(\hat{t}).$$

The test statistic T , is calculated as:

$$T = \frac{\bar{W} - \hat{w}_{\max}(\hat{t})}{S/\sqrt{n}} \sim t(n-1),$$

where S is the sample standard deviation:

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (W_i - \bar{W})^2}.$$

The p -value for the test is computed based on the t -distribution and quantifies the probability of observing a sample mean as extreme as $\bar{w}$ under H_0 . For a two-tailed test, the p -value is given by:

$$p = P(|T| > |t_0|),$$

where t_0 is the observed value of the test statistic.

Model Validation

To validate the age estimation framework, we utilize the same datasets as in the previous subsection. The expected maximum wear depth, or the expected minimum rise, is extracted from the simulated data for 100 years, and the sample means from both datasets are compared against this expected value.

1. Time-Depth Relationship

Figure 16 illustrates the monotonic decrease in the minimum rise over time, which serves as the basis for estimating the stairwell's age. The predicted depth for 100 years aligns closely with the sample mean of the first dataset.

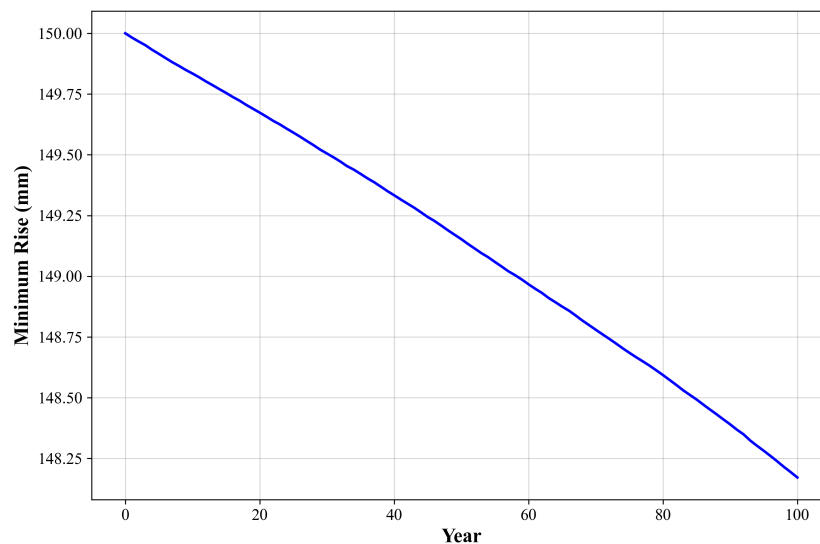


Figure 16: Relationship between the Minimum Rise and the Simulated Age

2. T-Test and Confidence Interval

A one-sample t -test is conducted for both datasets, comparing the sample means to the expected value for 100 years. The test statistic and confidence intervals are visualized in Figure 17. For the first dataset (Sample 1), the sample t -value falls within the 95% confidence interval, supporting consistency with the claim of 100 years.

3. P-Value Analysis

The calculated t -statistics and corresponding p -values are as follows:

- Sample 1 (100 years): $t = 1.364$, $p = 0.215$ (not significant, consistent with 100 years).
- Sample 2 (120 years): $t = -150.754$, $p = 0.000$ (significant, inconsistent with 100 years).

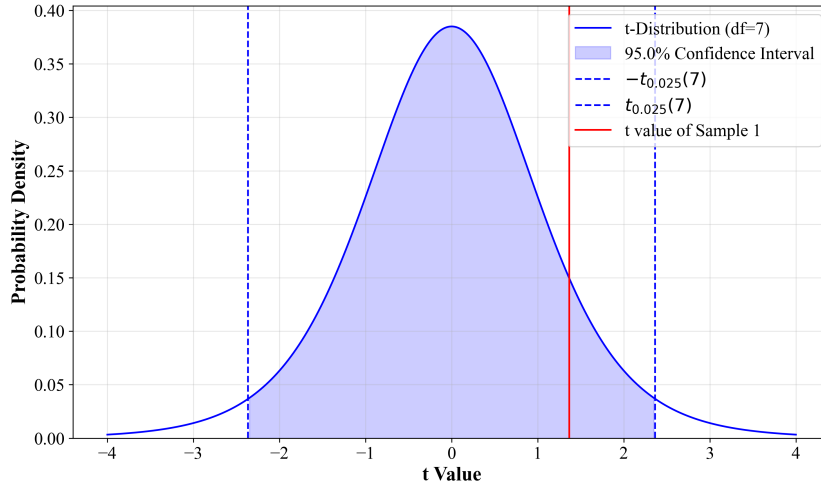


Figure 17: T-distribution with the Confidence Interval and the T-value of Sample 1

8.3 Detection of Repairs or Renovations

Understanding the structural history of a stairwell requires identifying sections that deviate significantly from the expected patterns of wear. Such deviations may indicate repairs, renovations, or irregular usage. This subsection proposes a methodology to detect these anomalies using a combination of **residual analysis** and **material uniformity testing**.

Model Development

To identify repaired or renovated sections, the measured height matrix H is compared with the predicted height matrix $\hat{H}$, derived from **Model I**. Deviations are quantified through residual analysis:

$$\Delta H = H - \hat{H},$$

where ΔH highlights areas where the wear depth deviates from the predicted patterns. Regions where $|\Delta H|$ exceeds a predefined threshold are flagged as potential anomalies.

Since the full H matrix cannot be directly measured due to practical constraints, a sampling strategy is employed to construct H . Discrete points are uniformly sampled within the worn regions of the stairwell to form the measured height matrix H . Corresponding points in the simulated matrix $\hat{H}$ are extracted to construct $\hat{H}$, ensuring consistency in comparison:

$$H = \{h_{i,j} \mid (i,j) \in S\}, \quad \hat{H} = \{\hat{h}_{i,j} \mid (i,j) \in S\},$$

where S represents the set of sampled locations. This approach allows the analysis to focus on representative wear patterns while remaining computationally efficient.

To further validate these anomalies, **non-destructive testing (NDT) using ultrasound** is employed. High-frequency ultrasound waves are transmitted into the material to measure parameters such as time of flight and attenuation. Repaired areas, often exhibiting distinct material properties, show deviations in acoustic responses. An anomaly is identified when:

$$\text{Anomaly} = \left| \frac{\text{Density}_{\text{measured}} - \text{Density}_{\text{expected}}}{\text{Density}_{\text{expected}}} \right| > \text{Threshold}.$$

In conclusion, the detection process comprises the following steps:

1. **Residual Analysis:** Construct the residual matrix ΔH using the sampled H and $\hat{H}$. Identify regions where $|\Delta H|$ exceeds a predefined threshold, flagging them as potential anomalies.
2. **Ultrasound Testing:** Perform localized ultrasound scans on flagged sections to measure material density and identify inconsistencies caused by repairs or renovations.
3. **Historical Cross-Reference:** Cross-check flagged regions with available historical renovation records to validate the findings.

This methodology leverages both statistical residual analysis and physical material testing to detect anomalies, ensuring comprehensive identification of repairs or renovations in a non-destructive manner.

8.4 Material Source Consistency

To Determine the consistency between the observed wear patterns and the presumed source material, this subsection presents a statistical model to evaluate whether the measured wear patterns align with the material properties of the presumed source, such as stone or wood.

Model Development

To evaluate the consistency between the observed wear patterns and the presumed source material, we compare the measured maximum wear depths of the stair treads with predictions generated using the wear coefficient of the presumed material. By incorporating the coefficient, along with other historical inputs such as age, daily traffic volume, and directional flow, into **Model I**, the predicted maximum wear depth $\hat{w}_{\max}$ is obtained.

For the measured data, the maximum wear depths $w_1, w_2, \dots, w_n$ of the sampled treads, defined in Section 8.2, are directly used. To determine if the source material is consistent with the observed wear, a **one-sample t-test** is performed. The null hypothesis H_0 assumes that the mean maximum wear depth of the measured data matches the predicted maximum wear depth:

$$H_0 : \mu = \hat{w}_{\max},$$

where μ represents the population mean maximum wear depth. The test statistic is computed as:

$$T = \frac{\bar{W} - \hat{w}_{\max}}{S/\sqrt{n}} \sim t(n-1),$$

where $\bar{W}$ and S are the sample mean and standard deviation, respectively, of the measured wear depths $w_1, w_2, \dots, w_n$.

The critical region for rejecting H_0 depends on the significance level α and the degrees of freedom $n - 1$. For a two-tailed test, the rejection region is:

$$|T| > t_{\alpha/2}(n-1),$$

where $t_{\alpha/2}(n-1)$ is the critical value from the t -distribution corresponding to a cumulative probability of $1 - \alpha/2$.

This approach allows archaeologists to statistically assess the compatibility between the observed wear and the presumed material source by leveraging measured wear patterns and simulated predictions.

8.5 Discussion

The proposed model serves as a robust framework for inferring historical usage and structural characteristics of stairwells by analyzing wear patterns through simulation, statistical validation, and non-destructive testing. It effectively evaluates wear consistency, estimates age, and detects anomalies but remains sensitive to sampling accuracy and assumptions about material properties.

9 Model V: Usage Pattern Inference Model

To distinguish between **short-term high-intensity** use and **long-term low-intensity** use of staircases, we develop a model based on **Maximum Likelihood Estimation (MLE)** that utilizes wear depth data to estimate the temporal decay parameter α . The observed wear depth, $S_{\text{measured}}(x, y)$, is assumed to follow a normal distribution centered around the predicted wear depth $S'(x, y, t)$, which is defined as Formula (5).

9.1 The Maximum Likelihood Estimation

The likelihood function is constructed based on the assumption that the observed wear depths are independent and normally distributed.

$$\mathcal{L}(\alpha, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(S_{\text{measured},i} - S'(x_i, y_i, t_i))^2}{2\sigma^2} \right).$$

By taking the logarithm, we derive the log-likelihood function:

$$\ln \mathcal{L}(\alpha, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (S_{\text{measured},i} - S'(x_i, y_i, t_i))^2,$$

where n is the total number of measured data points. Maximizing this log-likelihood function is equivalent to minimizing the mean squared error (MSE) between the observed and predicted wear depths. This allows us to estimate the key parameters α and σ^2 .

The temporal decay parameter α provides a direct indicator of the usage pattern. A larger value of α suggests rapid wear accumulation, which is consistent with short-term high-intensity use. Conversely, a smaller value of α indicates gradual wear over time, corresponding to long-term low-intensity use.

In addition to α , the edge wear discrepancy serves as a supplementary metric to further refine the analysis. The edge wear discrepancy is defined as:

$$\Delta S_{\text{edge}} = S_{\text{measured, edge}} - S'_{\text{edge}}(t).$$

A positive value of ΔS_{edge} suggests that the actual edge wear is greater than the predicted value, which aligns with long-term low-intensity use due to the accumulation of wear across the entire stair width. In contrast, a negative value indicates that the edge wear is less pronounced than predicted, which is characteristic of short-term high-intensity use, as wear is concentrated in the central regions of the stair treads.

9.2 Model Validation

We can validate the correctness of our conclusions through wear depth simulations. First, we need to set a fixed total wear amount and then input parameters corresponding to the two

different usage patterns. The results are shown in Figure 18 and Figure 19. From the Figure, it is evident that the edges of the stairs experience more severe wear under long-term low-intensity usage, which further confirms the validity of our model.

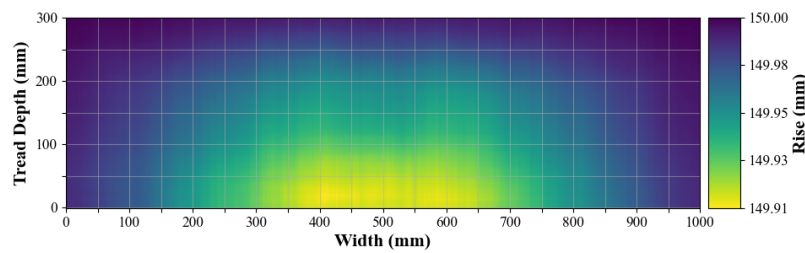


Figure 18: Long-term low-intensity pattern

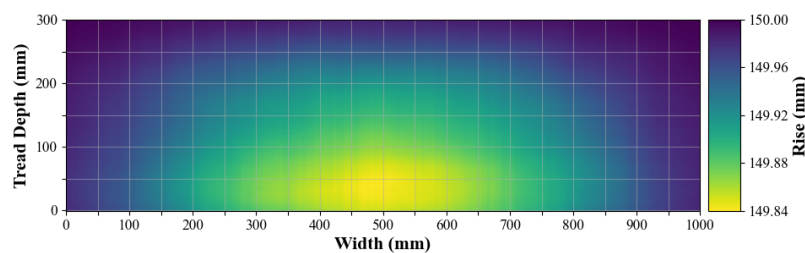


Figure 19: Short-term high-intensity pattern

10 Sensitivity Analysis

In order to analyze the impact of various factors on the wear depth simulation model, we have picked an important factor on this model: wear rate, since by our intuition, wear rate should present a linear pattern. The following figure shows the relationship between minimum wear depth (rise) and time:

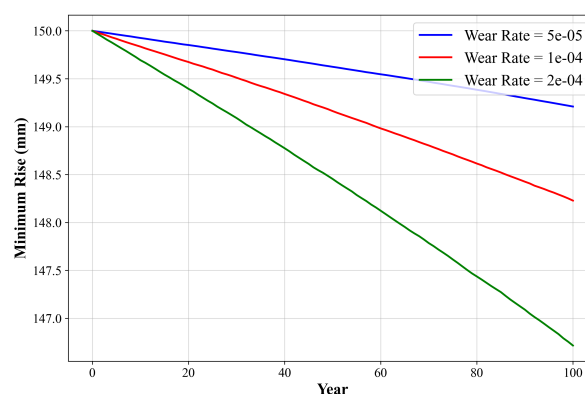


Figure 20: Minimum rise under different wear rates

From the visualization, it's clear that our model maintains a smooth prediction result, which is consistent with intuition, and demonstrates a relatively high degree of robustness against uncertainty.

11 Strength and Weakness

11.1 Strength

1. **Model is based on extensive research.** The model takes people's habits into consideration. For example, where do people usually place their foot and how much strength do they use when they ascend or descend stairs. Therefore, the model gives the ideal result of abrasion very close to reality.
2. **Model is highly extensible.** The idea can be used to further extend the model by adding more parameters to accommodate more complex situations, such as brick staircases, local climate and history.

11.2 Weakness

1. **Model lacks enough data support.** The model is based on the patterns of people ascending and descending stairs and the characteristics of worn stairs. However, there is not enough data to verify the accurateness, in particular, the relationship between wear of stairs and people's usage pattern. The validity of the model will be significantly improved if more accurate data can be obtained.
2. **Some factors are not considered by the model.** For example, people using the stairs might have been used to carry heavier things, which makes the estimation more inaccurate. Some ancient stairs were constructed using bricks, and the differences of abrasion between bricks and the spaces between them are ignored in our model. Local weather influence is also ignored, as rainfalls or sandstorms may accelerate the abrasion of some specific parts of the stairs. Therefore, there is a need to improve the model by taking more factors into consideration.

References

- [1] K. Fukushi, Y. Sekiguchi, K. Honda, H. Yaguchi and S. -I. Izumi, Three-dimensional GRF and CoP Estimation during Stair and Slope Ascent/Descent with Wearable IMUs and Foot Pressure Sensors, 2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Berlin, Germany, 2019, pp. 6401–6404, DOI: 10.1109/EMBC.2019.8856430.
- [2] Astrini Sie, Maxim Karrenbach, Charlie Fisher, Shawn Fisher, Nathaniel Wieck, Callysta Caraballo, Elisabeth Case, David Boe, Brittney Muir, Eric Rombokas, Descending 13 real world steps: A dataset and analysis of stair descent, Gait & Posture, Volume 92, 2022, pp. 383–393, ISSN 0966-6362, DOI: 10.1016/j.gaitpost.2021.10.039.
- [3] Li K W, Huang S Y, Chiu W H. Ground reaction force and required friction during stair ascent and descent[J]. Human Factors and Ergonomics in Manufacturing & Service Industries, 2017, 27(1): 66–73.
- [4] Robert Riener, Marco Rabuffetti, Carlo Frigo, Stair ascent and descent at different inclinations, Gait & Posture, Volume 15, Issue 1, 2002, pp. 32–44, ISSN 0966-6362, DOI: 10.1016/S0966-6362(01)00162-X.
- [5] Archard, J., 1953, Contact and Rubbing of Flat Surfaces, J. Appl. Phys., 24, pp. 981–988.

Report on Use of AI

1. OpenAI ChatGPT (Jan 28, 2025 version)

Query1: Can you provide detailed information on the materials and standard dimensions used in real-life staircase construction?

Output: Staircases are essential architectural elements that facilitate vertical movement within buildings. Their design involves careful consideration of materials and dimensions to ensure safety, functionality, and aesthetics.

Materials:

- **Wood:** Commonly used in residential settings, wood offers versatility and a warm aesthetic. Species like oak, maple, and cherry are popular choices due to their durability.
- **Concrete:** Favored in commercial and industrial applications, concrete provides strength and fire resistance. It's often used for both structural components and finishes.
- **Metal:** Steel and aluminum are utilized for their structural integrity and modern appearance. They are prevalent in contemporary designs and can be combined with other materials like glass.
- **Glass:** Used primarily for treads and balustrades in modern designs, glass imparts a sense of openness and light but requires careful engineering to ensure safety.

Standard Dimensions:

- **Tread Depth:** The horizontal part of the step where the foot rests. A minimum depth of 11 inches (28 cm) is standard.
- **Riser Height:** The vertical portion between treads. Heights between 4 to 7 inches (10-18 cm) are typical.
- **Stair Width:** For residential buildings, a minimum width of 36 inches (91 cm) is common, allowing comfortable passage.
- **Nosing:** The edge of the tread that protrudes over the riser below. Nosings can extend between 1 to 1.5 inches (2.5-3.8 cm), providing additional space for foot placement.

These dimensions aim to balance comfort, safety, and accessibility, accommodating a wide range of users.